An untargeted NMR metabolomics analysis of hippocampal regions of rats with SERT knockout gene.

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Abstract

There is a 50% reduction in serotonin function in humans suffering from depression and anxiety. Hence the current study employed an untargeted metabolomic analysis in serotonin knockout rats who mimic this 50% of reduction in serotonin transporter function of humans with depression. Untargeted metabolomics is an explorative analysis used to quantify metabolites in biological samples. A Nuclear Magnetic Resonance (NMR) technique was employed in this study to determine whether the quantity of any specific metabolic features varied significantly as a function of serotonin knockout genotype. During sample preparation, the samples had to be desiccated in two batches due to the number of samples that were able to run per time in the speed vacuum. Due to malfunctioning of the speed vacuum, there was a variation between samples of the two batches. Hence the two batches were analyzed separately leading to sample size within groups of batches to be very small. This resulted in a low statistical power hence leading to inability make any valid inference from the results. However, this study successfully employed sample preparation and optimization procedure for NMR analysis of rat brain tissue. In addition, it successfully employed an optimization procedure for pressing raw data to reduce potential artifacts that would impact results.

Introduction

According to Malhi and Mann, (2018), depression is a disease that was ranked as the third most debilitating disease in the world by the World Health Organization (WHO) in 2008. WHO predicts it will rise to the most debilitating disease by 2030. Diagnostic and Statistical Manual of Mental Disorders [DSM IV edition], and International Classification of Diseases [ICD] both define depression as a syndrome based on symptoms that impair normal functioning. According to DSM and ICD the diagnosis of Major Depressive Disorder (MDD) is based on persistence of five or more of the following symptoms: depressive mood or anhedonia or both present with fatigue, increased or reduced appetite, weight fluctuation, sleep disorders, lack of concentration and psychomotor retardation (Malhi & Mann, 2018). If the above symptoms last for two or more weeks the individual is diagnosed with depression (Malhi & Mann, 2018). A survey done in 2003 and 2004 to estimate the prevalence of DSM IV disorders for 20 months, by New Zealand Mental Health foundation indicated that 20% of adults reported mood disorders, anxiety, substance abuse or eating disorders (Scott et al., 2010). Furthermore, a higher prevalence of depression in females than males and the onset of the disease from adolescence to mid-40's was reported by Scott and al (2010).

Malhi and Mann (2018) indicated that genetic, biological, and environmental factors that lead to depression vary widely between patients. Therefore, intervention to treat depression should also differ from patient to patient. Hence, investigation of the neurobiology that underpins depressive symptoms is conducted to further understand the etiology and pathophysiology of depression and enhance intervention for depression (Malhi & Mann, 2018). MDD symptoms are linked to genetic and environmental factors and are associated with alterations in brain structures, in structural and functional connectivity and neurochemical deficits (Russo & Nestler, 2013). For example, neuroimaging studies indicate a reduction of volume in the prefrontal cortex (PFC) and hippocampus in MDD patients (Russo & Nestler, 2013).

The serotonin hypothesis of depression dates back to 1960s when Coppen shed light on the possible role of serotonin (5-HT). His reasoning was based on possible effect of antidepressants on 5-HT in regulating emotions in depressive patients (Craven & Bland., 2013). According to Coppen, drugs such as monoamine oxidase inhibitors decrease depressive symptoms and increase brain 5-HT levels (Craven & Bland., 2013). In addition, when large doses of amino acid tryptophan, the precursor for 5HT, are administered to patients, the antidepressive effects of monoamine oxidase-inhibitors are potentiated (Craven & Bland., 2013). The role of 5-HT in depression was further supported when administration of reserpine for hypertension led to both reductions in catecholamines and 5HT in the brain and increased severity of depressive symptoms (Craven & Bland., 2013).

A further important factor of the serotonergic system that has been linked to the development of MDD is the serotonin transporter (SERT). SERT is essential in transporting extracellular 5-HT back into the presynaptic neuron where it can be recycled for future use (Olivier et al., 2008). Information about the mechanism and function of the SERT is used in the development of medications for depression, such as selective-serotonin reuptake inhibitors (SSRI's). SSRIs inhibit the activity of the transporter, thereby increasing extracellular concentrations of 5-HT (Jans et al., 2007). A possible association between the serotonin transporter gene, *SLC6A4*, and depression has been hypothesized in numerous studies. These finding indicate that individuals with the less functional short allele of the serotonin-transporter-linked polymorphic region (5-HTTLPR) are more prone to stressful life events, which therefore predisposed to development of significant depressive symptoms (Caspi et al., 2003; Karg et al., 2011; Sharpley et al., 2014).Neuroimaging studies suggest that individuals with the short allelic 5-HTTLPR polymorphism exhibit greater amygdala hyperresponsiveness to negative environmental stimuli, encounter stronger adverse self-referential distress, and indicate higher activation in associated brain regions (i.e., dorsal anterior cingulate/dorsal medial prefrontal cortex and the right anterior cingulate) (Ma et al., 2013).

These findings point to a particularly interesting phenomenon generally referred to as the serotonin paradox. Thus, individuals with a genetically reduced SERT activity, leading to higher than normal extracellular levels of 5-HT are more prone to developing depressive symptoms. Yet on the other hand increasing extracellular levels of 5-HT by antidepressant drugs, such as selective serotonin reuptake inhibitor (SSRIs) like fluoxetine, reduce depressive symptoms. This paradox is further emphasized by the finding that fluoxetine administrated early in life in rats and mice induce symptoms of depression in adulthood (Oliver et al., 2011). One possible explanation for the 5-HT paradox is related to the timing of changes in 5-HT levels. Genetic and SSRIs treatment increase 5-HT levels early in life, while adults are administered with SSRI therapeutic treatment for depression. This idea is supported by a study by Rebello and colleagues (2014) who administer fluoxetine to postnatal day 2 -11 old rats. This treatment, during a critical developmental period in life, lead to changes in prefrontal cortex pyramidal neuron morphology which resulted in decrease excitability (Rebello et al., 2014). This is speculated to result in emotional deficits similar to anxiety and depression in adulthood (Rebello et al., 2014). Alterations in pyramidal cell morphology were also observed in cell cultures from rats with genetic reduction in SERT activity (Chaji et al., 2021). To analyze differences in dendritic spine density, primary cortical neurons cell cultures of wildtype ($SERT^{+/+}$) rats, Heterozygous ($SERT^{+/-}$) rats, and homozygous ($SERT^{-/-}$) rats immunocytochemistry (ICC) were examined. Specifically, neurons were stained for drebrin, a protein used to visualize dendric spines, and co-stained for microtubule-associated protein 2 (MAP2), which visualizes dendritic shafts (Chaji et al., 2021). The results indicated a gene dosage-dependent reduction in dendritic spine density in between $SERT^{+/+}$, $SERT^{+/-}$, and $SERT^{-/-}$. This study suggested that early 5-HT changes can lead to fundamental alterations in neuronal structure and connectivity, leading to an increased risk of depression later in life.

In line with the above-mentioned studies in humans linking the short allele of the 5-HTTLPR to depression, rats with a genetic reduction in SERT activity also show increased signs of depression and anxiety (Olivier et al in 2008). In addition, as expected, the extracellular serotonin levels of $SERT^{-/-}$ rats were significantly higher (Oliver et al., 2008). Furthermore, there was a reduction in 5HT and 5-hydroxyindoleacetic acid (5HIIA) tissue, a metabolite of 5-HT, in rats with the SERT mutation which suggests reduced levels of intracellular serotonin concentrations (Oliver et al., 2008). In addition, $SERT^{-/-}$ rats indicated 50% less serotonergic neurons in the DRN in comparison to mice (Oliver et al., 2008). Similarly, postmortem studies of depressive patients show similar reduction in serotonergic neurons in the DRN indicating that $SERT^{-/-}$ rats are a good animal model to study MDD in humans (Oliver et al., 2008).

All together these data support the idea that high levels of 5-HT early in life can lead to important and possibly widespread changes in brain connectivity, by affecting a multitude of neurodevelopmental processes. In the current thesis, we aim to investigate this by using a metabolomics approach to identify widespread biochemical changes in the brain (Coene et al., 2018). Metabolomics is a downstream process of genomics, transcriptomics and proteomics which help discover new information regarding various biological processors (Weng et al., 2015), and aims to identify metabolites and small molecular changes in an organism (Gonzales-Riano et al., 2016). There are two types of Metabolomic analysis: targeted biochemical assays and untargeted biochemical assays (Coene et al., 2018). While targeted metabolomics is used to analyze a limited number of known metabolites in a sample, untargeted metabolomics is employed to measure as many metabolites as possible, thus increasing the chance of identify novel biomarkers related to a disease of interest (Coene et al., 2018). Thus, it provides us an important tool to unravel metabolomic pathways related to certain central nervous system (CNS) diseases (Coene et al., 2018).

A metabolomic analysis carried out by Weng et al (2015) to identify the molecular mechanism in serotonin deficiency related to depression used both a targeted and untargeted approach using rat serum (Weng et al.,2015). A decrease in serotonin in the brain was achieved through knockout of Tph2 in rats of 12 and 16 weeks of age (Weng et al.,2015). This yielded 17 metabolites that are related to serotonin deficiency (Weng et al.,2015). This metabolomic study indicated competitive pathways for tryptophan where inhibition of one resulted activation of the other (Weng et al.,2015). That is kynurenine and serotonin are both synthesized from tryptophan (Weng et al.,2015). Therefore, the activation of kynurenine pathway results in a deactivation in the serotonin pathway, of which the deficiency is concomitant with depression (Weng et al., 2015). Furthermore, there is an increase in indolemine2,3-dioxygenase an enzyme that metabolizes tryptophan to kynurenine in depressed patients (Oxenkrug., 2013). This leads to an increase level of kynurenine in the cerebrospinal fluids of depressed patients (Oxenkrug., 2013). Therefore, when diagnosing MDD and its severity the kynurenine pathway is taken into consideration (Oxenkrug., 2013). In addition, due to the mismatch in antidepressants that lead to increase in energy levels relative to mood improvement which results in suicide during treatment suggests, a treatment that modulates Kynurenine pathway might lead to better outcomes in depressive patients (Oxenkrug., 2013).

In our lab, an untargeted metabolomic study was done by Henry Chafee (2020) using hippocampal tissue samples of the SERT$^{+/+}$, SERT $^{+/-}$, and SERT$^{-/-}$. rats to investigate global biomarkers and metabolomic difference to potentially discover new pathways linked to depression. After optimizing the sample preparation protocol Liquid Chromatography/Mass Spectrometry technique (LC/MS) was used to analyse the hippocampal samples. Unfortunately, due to high usage of the LC-MS machines by different schools the samples had to be divided into 2 batches for both the male and the female rats. Because of periodic recalibration and maintenance of the machines in between the runs, the study suffered from a highly significant batch effect. This resulted in a significant reduction in the statistical power preventing us from identifying significant difference between the genotypes.

The current study aims to follow on from the study done by Chafee (2020), employing nuclear magnetic resonance (NMR) instead of LC/MS to detect metabolites in male and female SERT$^{+/+}$, SERT $^{+/-}$, and SERT$^{-/-}$. rats. LC/MS is more sensitive than NMR, with lower limits of recognition usually being 10 to 100 times better and, therefore, it has been used by more than 80% published research (Emwas et al., 2019). However, in recent years NMR metabolomic studies are on the increase (Takis., 2019). This is due to the advantages on NMR technique over LC/MS and other analytical technique (Guennec et al., 2014). Some of these advantages of NMR spectroscopy are its nondestructive, nonbiased nature, and little or no necessity for chromatographic separation, sample treatment, or chemical derivatization (Guennec et al., 2014). Furthermore, it permits simple quantifiability and routine detection of new compounds and can be operated automatically, it is highly reproducible and has a high throughput (Guennec et al., 2014). Due to these advantages, the current study will employ exploratory metabolomic study using NMR platform to identify whether there is a significant difference in metabolite concentration in rats with a genetic reduction in SERT functioning.

Materials and Methods

Animals

The hippocampal tissue of 42 animals (22 females and 20 males) was examined in the current study. In the female group, 8 were $SERT^{+/+}$, 6 were SERT $^{+/-}$, and 8 were $SERT^{-/-}$. In the male group, 4 were $SERT^{+/+}$, 8 were $^{+/-}$, and 8 were SERT $SERT^{-/-}$. All animals were reared in the Victoria University of Wellington vivarium and sheltered in single ventilated polycarbonate cages in a room kept on a 12-hour light/dark cycle (lights on at 7PM); ambient temperature was retained at 21°C and humidity at 55%.

On the day of sacrifice, animals were euthanized with carbon dioxide gas, and their brains immediately removed and instantly frozen in isopentane with dry ice at -35°C preceding to storage at -80°C. At the time of sacrifice, animals were adults and varied in age from 55-133 days, with the majority between the ages of 55 and 59; 5 females (2 $SERT^{+/+}$ and 3 SERT $^{+/-}$), and 2 males (both SERT $^{+/-}$), were significantly older (97-133 days). These 7 samples were eventually eliminated from the final statistical analysis to avoid a potential confounding effect of age. All processes were assessed and approved by the Victoria University of Wellington Animal Ethics Committee.

Sample Preparation

Tissue Dissection

The hippocampus was dissected manually from the previously frozen brains. Briefly, thawed tissue was placed on a glass plate set in crushed ice, and the hippocampus was rapidly isolated and removed free hand. Using Heffner et al. (1980) as a reference for the coronal cuts, the tissue was sliced, and a 1.5 mm slice was cut with cold scalpel blades. The left hippocampus was then obtained by separating it from the cortex.

Preparation Protocol Optimization: Brain Tissue

In order to optimize signal detection of molecules present in the brain tissue, we first needed to optimize the sample preparation method. To that end, we compared acetonitrile (ACN) and methanol (MeOH) preparation followed by partial desiccation using Nitrogen flow, and speed vacuum to test the evaporation of organic solvents after extraction. For ACN preparation we combined each 1 mg of tissue with 10uL of 50% ACN using a 5 mL tube for centrifugation and 1.5 mL tube to homogenisation with a conical pestle for about 1 minute. For MeOH preparation the same procedure was used by replacing ACN with MeOH. We vortexed the solvent for 10-30 seconds and centrifuge at 15000 g for 15 minutes at 4°C. All the supernatant was collected, and 50% D2O with DSS (sodium trimethylsilylpropane-sulfonate) was added at a ratio of 2:1. D2O was used as an internal standard to stabilize signals or minimize signal drift. For desiccation of the ACN solvent, the tissue sample was mixed with ACN, and supernatant was obtained at a 1:10:10 ratio. For Desiccation of MeOH the same procedure was employed. Both ACN and MeOH solvents were evaporated using speed vacuum overnight. As all 42 samples could not be speed vacuumed at once, the desiccation process required two batches. However, due to the speed vacuum malfunctioning, batch 2 was not desiccated overnight in the speed vacuum. In addition, after desiccating the samples of that batch again, the temperature of the samples was slightly high, which could have induced some possible degradation during desiccation. The desiccated samples were stored at -80°C until ^{1}H NMR analysis. The desiccated samples (ACN and MeOH) were reconstituted with D2O and DSS solvent to give the same volume as before desiccation.

The reconstituted samples were then transferred to 5-mm NMR tubes for 1H NMR analysis. The partially desiccated solvents had lesser peaks in both ACN and MeOH solvents in comparison to fully desiccated solvents with the speed vacuum. In addition, the desiccated ACN solvent had marginally more peaks, but the peaks were split whereas MeOH solvent led to higher peaks. Further comparison of both signals to previous literature indicated that the MeOH preparation led to results similar to those obtained in the literature. As a results, we decided to use MeOH in our final preparation protocol. During this period of protocol refinement, we also determined to use sodium trimethylsilylpropane-sulfonate (DSS) as the internal standard in our samples. ISs are standard, non-naturally occurring substances that are frequently included in metabolomic analytical protocols to improve the accuracy and precision of quantification methods (Tan & Awaiye, 2013). First, we tested to see whether the IS would be visible by running an NMR analysis with D2O and DSS solvents combined. This also indicated the water signal which was suppressed when analysing NMR signals for both (ACN and MEOH) samples.

Final Preparation Protocol

With the completion of these optimization tests, we finalized the protocol to use for all our biological samples. After dissection of the brain and isolation of the hippocampus (Figure 1), we weighed a sample of the region (typically between 35 mg to 65 mg) and combined each 1 mg of tissue with 10 uL of 50% MeOH. We used a 5 mL centrifuge tube and homogenized the tissue with a conical pestle for about 1min (or until visually homogenized), transferred the homogenate to a 1.5 mL microcentrifuge tube, and centrifuged at 15000 g for 15 minutes at 4° C. Then 300 uL of supernatant (SNT) was collected to desiccate in the speed vacuum overnight. Afterwards, we stored the desiccated sample at -80°C until analysed. On the day of, and just before the ^{1}H NMR analysis, to reconstitute the sample we added 600 uL of D20/DSS and vortex for a few seconds, and transfer to a 5-mm NMR tubes.

Just before ^{1}H NMR analysis we added 600 uL of D2O/DSS instead of 550 uL because we needed at least 500 uL of final volume in the NMR tubes, and for some of the samples SNT collected was less than 300 uL (285 and 250 uL), so adding the proportional volume of D2O/DSS would have led to total sample volumes less than 500 uL. Thus, by adding 600 uL, the lower SNT samples ended up with a volume of 570 and 500 uL, respectively. When adding the D2O/DSS, some of the desiccated sample didn't resolubilise at all and were floating in the sample, preventing the NMR analysis to be conducted properly. This could have been due to non-solubilising parts of the pellet being drawn through the pipette when collecting the SNT. To have clear samples, Micron RC-3000 membrane spin filters (15000 g for 5 minutes at 4° C) were used to collected both filtered and unfiltered solution. The filtered samples were then transferred to 5-mm NMR tubes.

Nuclear Magnetic Resonance (NMR) Spectrometry

NMR spectroscopy plays vital and multidimensional functions that have helped and continue to help the field of metabolomics. These include biomarker discovery (Emwas et al., 2013) kinetic and mechanistic metabolic experiments metabolite imaging (Corrias et al., 2018) food and beverage classification (Radursoga et al.,2017), environmental monitoring (Garcia-Sevillano et al.,2015) and drug toxicity research (Dunn et al., 2011). The unique features of NMR which are high level of reproducibility, its ease in sample preparation, its capacity to handle diverse sample types (liquids, solids, gels), its quantitative abilities, and its efficacy in recognizing unidentified metabolites along with its nondestructive nature, have made it predominantly beneficial in many varied metabolomic disciplines (Emwas et al., 2019). The near total computerization in the NMR workflow (with sample preparation, sample loading, spectral acquisition, and spectral deconvolution) is another reason why it is widely used in many large-scale metabolomic investigations (Emwas et al., 2019). While low sensitivity is considered as an inherent limitation of NMR spectroscopy compared to mass spectrometry, enhancements in probe design, magnet design, magnet field strength, pulse sequences, and methods for sensitivity improvement are resulting in NMR techniques been used widely in metabolomic studies (Emwas et al., 2019).

We conducted our sample analysis using a 600 MHz supercool Cryogenic probe. A normal proton analysis was employed using a Watergate pulse sequence to suppress the water signal. The Watergate pulse sequence technique which uses plus field gradients and is independent of the line shape, yields better suppression compared with other water suppression methods. Exchangeable protons are not affected and there is no phase jump at the water resonance, although signals the are close to the water resonance were suppressed. The signals of each sample were obtained within the range of -0.335 ppm and 9.665 ppm. The water signal was suppressed at 4.6652 ppm. 128 scans were acquired with a relaxation delay of 3 seconds. The resolution of the signals was adjusted to 32768 points and the spectrums were referenced to the known concentration of the added DSS (IS) peak, at 0 ppm.

Data processing protocol

Raw data was imported to MestreNova and an NMR spectrum for each sample was obtained. Processing of NMR data consisted of four steps: Fourier transformation, phase correction, baseline correction and calibration (Goodacre et al., 2007). Fourier transformation was used to transforms the raw real-time data into the frequency domain (Goodacre et al., 2007). Phase correction was applied to corrects the phase of the resulting NMR spectrum (Goodacre et al., 2007). Baseline correction was employed to ensure a constant zero baseline across all NMR spectrums (Goodacre et al., 2007). This reduces offset of the intensity values in order to acquire accurate peak alignment and quantification (Goodacre et al., 2007). Calibration was done to ensure that chemical shift scaled across all spectra was consistent, using the DSS internal standard (Goodacre et al., 2007). Chemical shift calibration ensures consistent global alignment of spectra in a metabolomics data set which is critical for statistical analysis (Goodacre et al., 2007). The correct spectral alignment is required for reliable metabolite identification (Anderson et al., 2010).

The data within the range of -0.36ppm and 5ppm was selected to be processed in Metaboanalyst as most of the data were present within this range. Binning is used to reduce data into small segments of equal width (Wishart, 2008). The full resolution of the data was reduced to bin width of 0.04ppm. In addition, data were normalized to the internal standard (Akira et al., 2008) DSS (0.004ppm). The samples were assessed separately as desiccation took place in two batches. Thus batch 1 and batch 2 were analyzed to identify whether there was a significant difference between the two batches.

Statistical analysis was performed using MetaboAnalyst, an online web server that utilizes R code for the processing and statistical analysis of metabolomics data sets (Pang et al., 2020;). The data within the range of -0.36ppm and 5ppm was selected to be processed by MetaboAnalyst as most of the data were present within this region. The program was used to normalize biological samples to the internal standard DSS (0.004ppm). The data was scaled using the pareto scaling function to permit multivariate analysis. These procedures were employed from previously established recommended measures for analyzing and interpreting metabolomics data (Broadhurst et al., 2006). Similar to the final preparation protocol, this processing was the result of comprehensive optimization efforts; for a complete and graphical explanation, please refer to the appendix A.

Statistical Analysis

Statistical analysis was performed using MetaboAnalyst. At first, a principal components analysis (PCA) was performed in order to decide the extent of overall group separation and the metabolites that led to that separation. PCA is a non-parametric, unsupervised statistical analysis which permits for recognition and visualization of the metabolites that lead to most considerably to variance between samples (Nyamundanda et al., 2010). Nevertheless, since it is unsupervised, it does not allow direct assessment of metabolites that result in split between groups of samples.

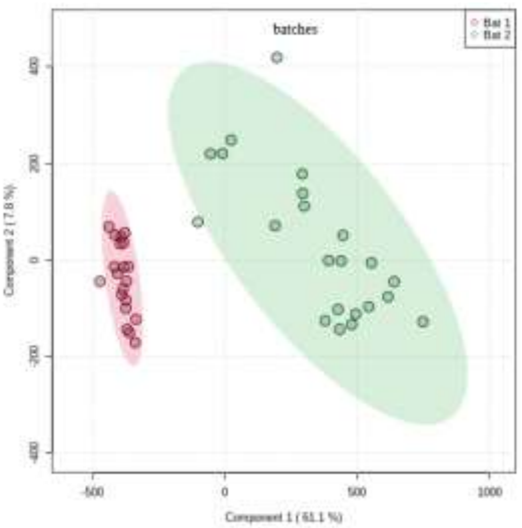
Thus, to directly analysis for possible variations amongst genotypes, a partial least squares-discriminative analysis (PLS-DA) was employed. PLS-DA is a supervised, multivariate regression analysis that is frequently applied to categorize and identify possible biomarkers from metabolomics data sets (Szymańska et al., 2011). A post-hoc test of validation procedures, consist of double cross validation and permutation testing, were employed to make sure that the PLS-DA model was adequately optimized to allow for significant inferences to be made from the results (Szymańska et al., 2011). Validation measures (R^2 value and Q^2 value) were applied to verify whether the models met goodness-of-fit criteria to be considered valid. The Q^2 is a default diagnostic statistic that is applied to assess the degree of error between the predicted categorical variable and the known variable (Szymańska et al., 2011). The quality of the prediction model can be measured by R^2. It is the fraction of variance of the response that is explained for the calibration model samples. R^2 indicates how well the model predicts the calibration of data. If the quality of a given PLS-DA model is moderate or even weak the model has little predictive power. This is reflected in the difference between R^2 and Q^2. If the difference between R^2 and Q^2 is too large (greater than 0.03) the scores plot from calibration models will not give meaningful visualization of pertinent information in the model (Bevlacqua & Bro, 2020).

Permutation testing includes the allocation of samples to random classification systems to decide whether random models clarify the variance clearer than the constructed model. The test yields a p-value, and the model is regard as statistically robust if $p < 0.05$ (Szymańska et al., 2011). Permutation testing includes the allocation of samples to random classification systems to decide whether random models clarify the variance clearer than the constructed model. If the test yields a p-value lesser than 0.05 ($p < 0.05$) then the model is regard as statistically robust (Szymańska et al., 2011).

Results

While all samples were prepared on the same day to minimize variation between samples they were not desiccated on the same day as only 22 samples fit in in the speed vacuum at a time. Thus, the whole sample was divided into two batches to be desiccated in the speed vacuum. However, while batch 1 was desiccated overnight batch 2 was not desiccated overnight as planned. Hence batch 2 was left in the speed vacuum for another extra day. Therefore, we first performed a PCA analysis on the two batches to identify whether there was a difference between the two desiccation procedures. As indicated in figure 1 the two baches clustered into two separate groups. Batch 1 showed a very close

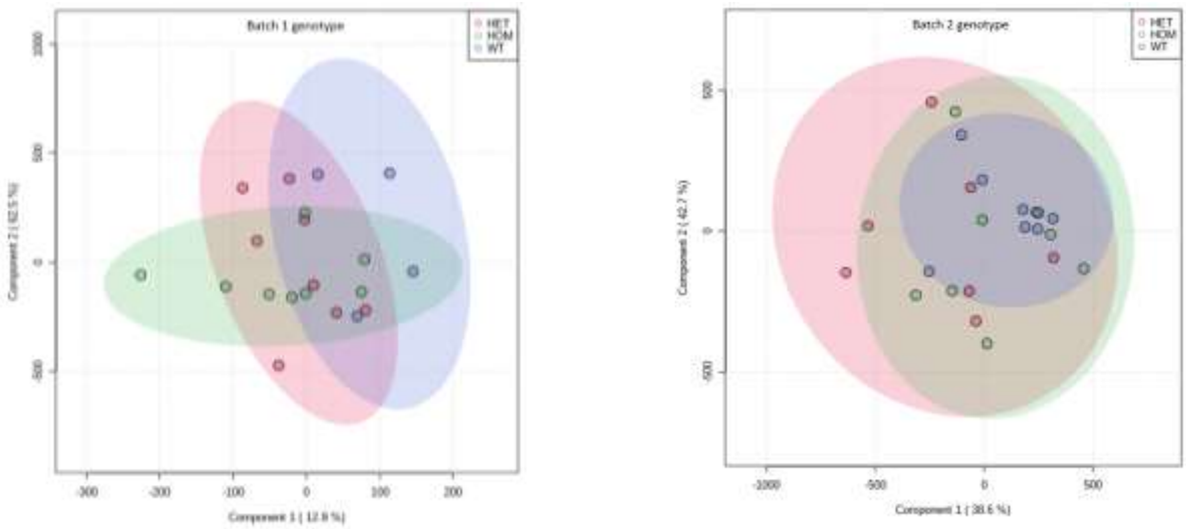
Fig 1.



clustering between samples implying less variation, while batch 2 showed a larger variance. However, PCAs are only a preliminary statistical visualization technique. To verify whether groups of samples varied significantly as a function of metabolite features a partial least square-discriminative analysis (PLS-DA) regression model was employed. As indicted in table 1 the Q^2 and R^2 for batch differences were identical indicating the visualization in score plots from the calibration models provided valid information. Thus, it could be inferred that the PCA indicated pertinent information about difference in the two batches. In addition, the permutation p value for the batches reached a significant level (p <0.001) from 1000 permutations. Overall, this shows that there were significant differences between the two batches and therefore that preparation had a significant influence on the metabolite profiles.

As there was a significant difference between the two batches, the batches were analyzed separately for genotype and sex. The PCA analysis as seen in figure 2 shows that for batch 1 there was a certain degree of genotype dependent segregation, while there was substantially more overlap between the genotypes in batch 2.

Fig 2.

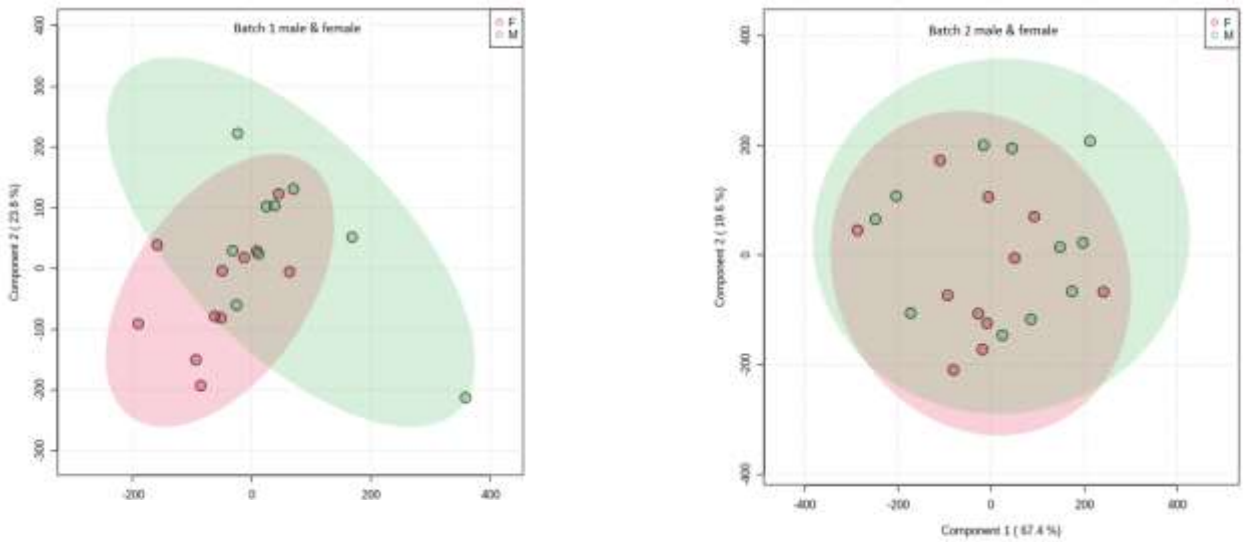


However, Q2 and R2 for batch 1 difference was 0.6 which is greater than 0.03, indicating that the visualization in score plots from the calibration models did not indicate predictive power as the difference is greater than 0.03. The difference between Q2 and R2 in batch 2 genotype was zero, which is lesser than 0.03 indicating good predictive power. However, the permutation p value was non-significant for batch 1 and batch 2 genotype (p=0.720 and p=

0.414 respectively). This indicates the difference observed in genotypes in both batches were not significant.

In addition, an analysis was done to examine whether there was a difference between metabolite profiles in the two sexes in the two baches. As indicated in figure 3 while batch 1 indicated some variability amongst metabolites between genders batch 2 showed an overlap of metabolite components between males and females. The difference between Q2 and R2 in

Fig 3.



batch 1 were 0 which is lesser than 0.03, indicating predictive power. However, the difference between Q2 and R2 were larger for batch 2 which is 0.89 indicating weaker predictive power. Nevertheless, the permutation p value was non-significant in batch 1 and batch 2 (*p=0.726 and p=*0.285 respectively).

Moreover, the genotypes for each gender in each batch were analyzed. However, since the sample size was very small any inference at this point in time would be merely speculative. For graphical explanation of these analysis please refer to the appendix A.

(Table 1).

Group	Q^2	R^2	Permutation

Batch differences	Q^2 =0.98	R^2 = 0.98	$p < 0.001$
Batch 1 genotype	Q^2 = -0.30	R^2 =0.30	p=0.720
Batch 2 genotype	Q^2 = 0.02	R^2 = 0.02	p= 0.414
Batch 1 male female differences	Q^2 = -0.3	R^2 =-0.3	p=*0.726*
Batch 2 male female differences	Q^2 =-0.32	R^2 =0.57	p=0.285
Batch 1 female genotype	Q^2 = -0.4	R^2 =3.7	p=0.994
Batch 2 female genotype	Q^2 = 0.6	R^2 =0.9	p=0.435
Batch 1 male genotype	Q^2 = 0.06	R^2 =0.98	p=0.094
Batch 2 male genotype	Q^2 = -0.47	R^2 = 0.78	p=0.936

Discussion

The aim of the current study was to use NMR untargeted metabolomic analysis to identify quantitative differences in metabolites in genotypes that would indicate possible biomarkers or metabolic pathways that are altered in animals with reduced SERT functioning. In order to optimize signal detection in samples several data processing protocols were employed to the raw data.

Initially the organics extracted, which are lipids and potential metabolites were dried under N2 flow until the MeOH solvent was evaporated before adding D2O/DSS prior to NMR analysis. However, this took a long time which could potentially lead to degradation of samples due to been outside the freezer even though it was placed on ice throughout desiccation. In addition, whether or not the solvent was evaporated was decided on whether the experimenter was able to see any remaining solvent or not. Hence, it was hard to determine whether the solvent was completely evaporated or not merely by looking at it. In order to optimize peak detection, the solvent needed to be completely evaporated. Hence, we decided to evaporate the solvents in the speed vacuum. However, only 22 samples were able to be evaporated at a time in the speed vacuum. Hence, we decided to evaporate the samples in two batches. For complete evaporation of solvents, it needs to be in the speed vacuum for 24 hours. The first batch was successfully evaporated in the speed vacuum overnight for 24 hours. However, the second batch was not evaporated overnight as expected. This was due to some sort of mechanical failure in the speed vacuum which is further discussed under limitations. Thus, the second batch had to be left in the speed vacuum for another 24 hours.

Hence, we decided to analyze the two batches to examine whether preparation might have led to metabolite profile differences between the two batches. As speculated the PCA analysis indicated a graphical difference in components which suggested possible batch effects. In addition, the PLS-DA indicated a significant difference between the two batches. Therefore, the two batches were analyzed separately to eliminate this confound. The PCA analysis of the two batches indicated that metabolites in batch 2 showed higher variability whereas batch 1 indicated very little variability resulting it to cluster together. In addition, the following PCA indicated a visible difference between genotype alone and gender in batch 1. In contrast batch 2 indicated no such differences between genotype or gender. However, this could have been a result of less variability amongst components in batch 1 that might have highlighted the slights differences in metabolites opposed to batch 2. Nevertheless, the PLS-DA analysis only indicated a significant difference between the variance in the two batches and not groups within batches. Hence, we cannot conclude that the difference between genotype and gender observed in batch 1 was significant.

Following analysis, several iterations of data processing protocols were applied to the raw data, in order to reduce any confounds due to batch effects. Despite these efforts, the study failed to find significant differences in metabolite concentrations between genotype groups, largely as a result of the speed vacuum malfunctioning, that will be discussed in the limitations section. As expected, better dispersion of genotypes was observed between the groups (genotype and gender) of batch 1 than batch 2. Samples clustered together in batch one might must have led to a separation of genotypes and gender within batch one as observed in PCA score plots. However, the statistical models still did not reach the level of significance in further analysis within batch one. The degradation speculated to have occurred in batch 2 due to failure in the speed vacuum to operate properly might have led to the cluster to be spread out and genotype not indicating variability within the batch.

However, having to separate the 42 samples into two batches led to batch one having 22 samples and batch two having 20 samples. Further separation of batches based on genotype, gender differences, genotype based on gender differences made each sample size to be extremely small. This sometimes led a genotype within a batch having only 4 samples. This is evident in batch 1 genotype with the only four samples representing wild types. Hence this small sample size could have led to reduced statistical power.

Strengths

The meticulous procedure of improving and optimizing the sample preparation technique produced a vigorous protocol. Through various pilot studies we optimized the protocol of sample preparation to yield a comprehensive metabolic profile while reducing extraneous noise. Both MeOH and ACN were used in the pilot study to determine which solvent would yield the greatest number of metabolite peaks. In order to reduce variability due to sample preparation and storing time which could potentially lead to degradation all tissue samples were extracted and SNT was collected and stored together to be desiccated. The desiccation of the solvent was done with the speed vacuum instead of N2 to make sure the solvent was completely desiccated and would not interfere with peak detection. In addition, all samples were prepared, and NMR analysis was employed on a single day. Since only 22 samples fit into the speed vacuumed a statistical analysis was employed to examine whether there were any batch differences to reduce any variability that would lead to misleading results. Once significant differences were detected the two batches were analyzed separately to reduce confounds impacting the results.

Limitations

Due to the limited number of samples that could be run at a time in the speed vacuum the 42 samples were divided into two batches for desiccation. Hence, we could not reduce possible degradation or variability that would result due to different time lengths different samples were stored in the freezer prior to desiccation and after desiccation till NMR analysis. In addition, samples of batch two were not desiccated within 24 hours and had to be left in the speed vacuum for another 24 hours. Usually, the speed vacuum offers the capacity to regulate chamber temperature pre- and post-sample run to prevent thermal degradation of heat sensitive samples such as biomolecules. Thus, there could be a possibility that the malfunctioning in the speed vacuum must have also resulted a failure in temperature regulation. If this was the case, then this would have led the samples of batch two to result in change in biomolecules due to degradation. In addition, when the samples of batch 2 were removed from the speed vacuum it indicated a higher temperature than usual. This may have been the reason that the metabolites in samples of batch 2 might have indicated a high variability as seen in the PCA graphs in batch differences. This led us to analyses the samples in two groups which led to the sample size to be very small. This might have led to low statistical power and possibly led to a false negative result. Hence this might have been the reason why even though batch 1 indicated close clustering the following statistical analysis did not lead to a significant difference between genotypes, or genotypes between gender differences. Due to time constrains we were not able to reprepare the second batch which would have possibly lessened variability between batches, hence address the problem of statistical power by statistically analyzing all 42 samples together. This would have resulted a big enough sample even when genotype between gender differences is performed and resulted in a possible significant difference.

Future directions

There are various measures future research could undertake in order to reduce batch effects that would lead to an increase likelihood of significant findings. Informed through this study of possible speed vacuum malfunctioning the future researchers could structure the time frame of the study to amend for such a draw back. That is, if they come across an obvious degradation of samples due to mechanical malfunctioning as encountered in this study, they could reprepare the samples of that particular batch by extracting hippocampal tissues of preserved rat brains. While this would slightly lead to differences between batches due to preparation times it would not be as extensive as the degradation the sample undergoes as a result of possibly left in room temperature for duration of time as may have occurred due to the speed vacuum malfunctioning. Alternatively, they could prepare all 42 samples again to reduces variability that preparation time might pose.

In addition, while the current study incorporated NMR even if we got significant results, we might have not detected all possible metabolites present in genotypes due to low sensitivity. Hence, incorporating a GC/MS would address both the issues posed by Henry Chaffer (2020) study using LC-MS and the current study using NMR. That is GC/MS has high sensitivity unlike NMR but also has a high throughput unlike LC/MS (Bhinderwala et al., 2018). Furthermore, studies indicate incorporating both GC/MS and NMR together in metabolomics studies are highly complementary (Bhinderwala et al., 2018). Bhinderwala and colleagues (2018), indicate that the different metabolite profiles were identified by NMR an GC/MS leading to an overall high detection and annotation of metabolites. Since, combining of techniques are possible through similar preparation methods (Bhinderwala et al., 2018) a future study based on these two techniques will be able to overcome possible drawbacks of

NMR and LC/MS alone. In addition, using two techniques could also lead to some valid results in case of a mechanical malfunctioning in one technique.

Conclusion

This study was a continuation of Henry Chafee's LC/MS untargeted metabolomic analysis of the Serotonin Knockout Rat Hippocampus. The current study employed an untargeted metabolomic analysis to identify possible significant differences in metabolite profiles that would have resulted due to disruption of serotonin transporter functioning. If we were able to successfully detect such difference this would have led to the identification of a potential biomarker for symptomology exhibited by depression and anxiety in SERT KO rats. This would have resulted in identifying the metabolic pathway that is mostly impacted by such disruption. In order to identify these pathways, the hippocampi of HOM, HET, and WT animals were extracted and homogenized in MeOh, and analyzed using NMR technique. Raw data was extracted and processed using MestreNova and statistical analysis was done with MetaboAnalyst software. Finally, despite efforts of considerable optimization and refinement in both sample preparation and processing of statistical analysis we did not discover any individual metabolite features that varied across genotypes. Future research could be aware of possible drawbacks due to mechanical malfunctioning and organize the time frame to have to go back and prepare tissue samples if needed, to optimize detection of metabolite profiles. In addition, combining two techniques such as NMR with GC/MS would also provide some compensation in case of such mechanical malfunctioning that would lead to time constraints resulting in failure to reprepare samples. Despite the limitation, future metabolomic projects will benefit from this study's considerable contribution to optimizing sample preparation and data processing protocols.

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Appendix A

Processing Optimization process

In order to obtain significant results, initially both the full spectrum range, and range between-0.36ppm & 5ppm with bin width varying from 0.01 to 0.06 was selected in MestreNova for statistical analysis in MetaboAnalyst and compared. Finally, data within the range of -0.36ppm and 5ppm was selected to be processed in Metaboanalyst as most of the data were present within this region. The normalizing function, “to normalize biological samples to the internal standard DSS” 0.004ppm was selected. The data was scaled using the pareto scaling function to permit multivariate analysis. These procedures were employed from previously established recommended measures for analyzing and interpreting metabolomics data (Figure 1). Due to variation in sample preparation the samples were divided into two batches for statistical analysis using MetaboAnalysit software (Figure 2).

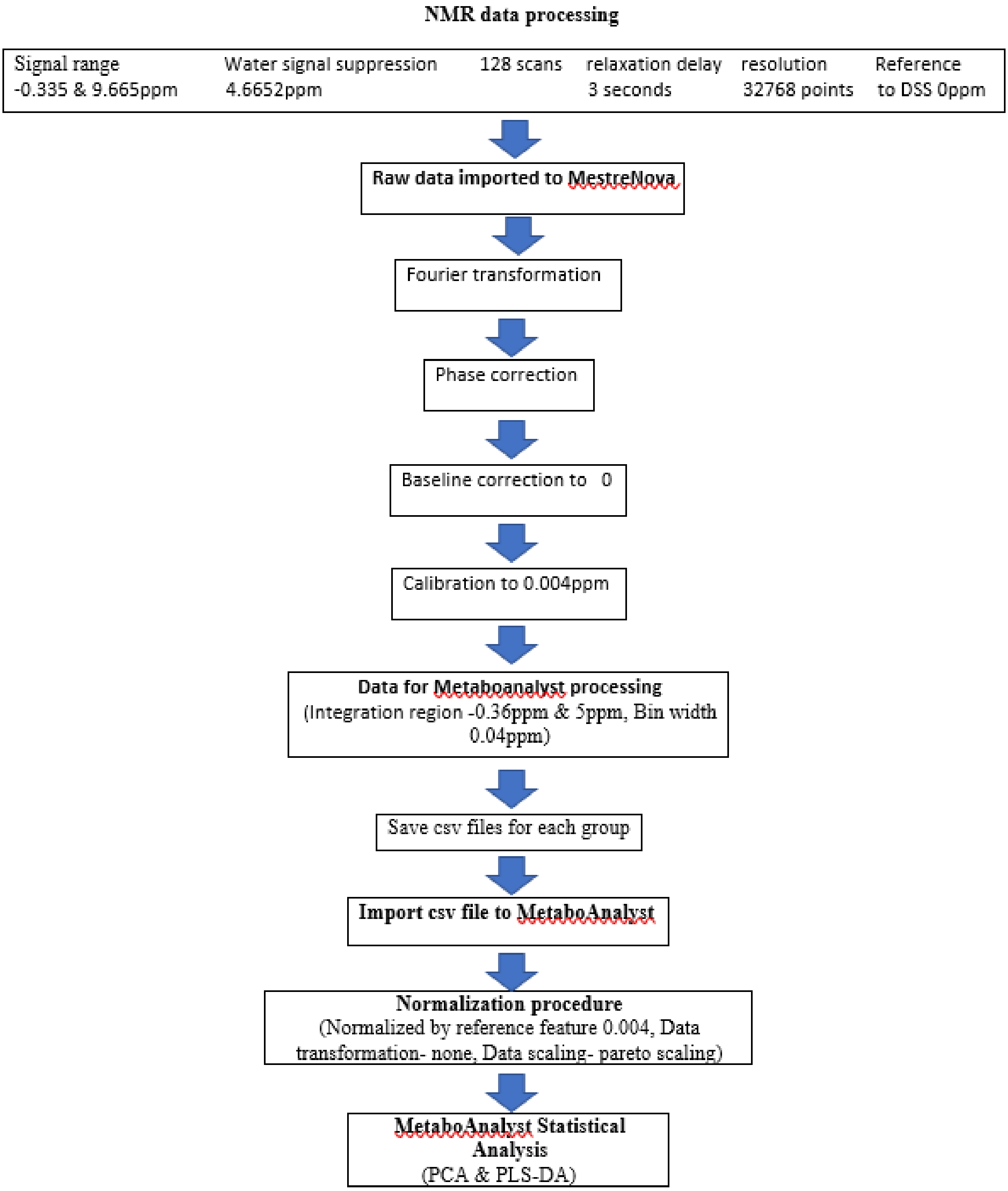

Figure 1. Data processing workflow

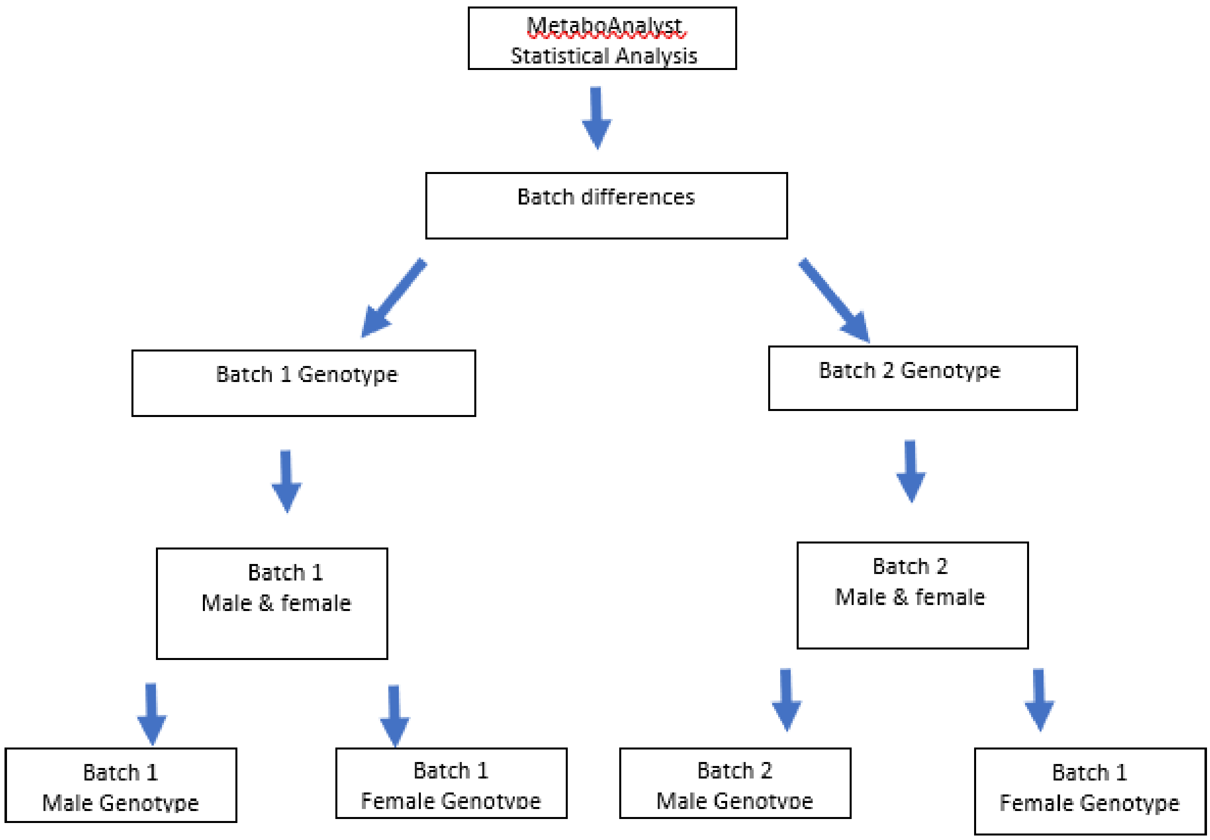

Figure 2. Statistical analysis for different groups.

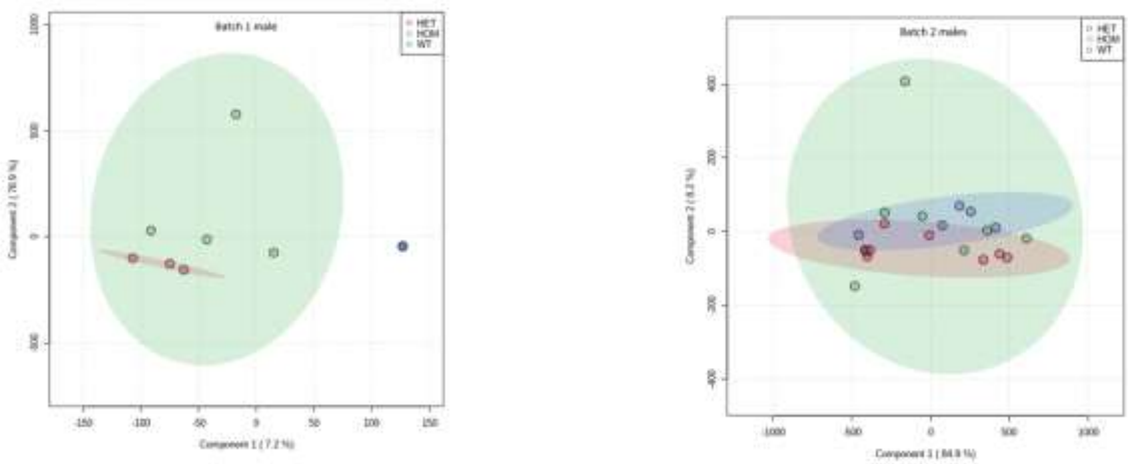

Figure 3. PCA analysis for male genotype in batch 1 and batch 2

The PCA for batch 1 male indicated a difference in genotypes but batch 2 males indicate an overlap of genotypes. This could be speculated to be due to close clustering of the samples in batch 1 which would have led to the slightest variation present to be highlighted (Figure 3).

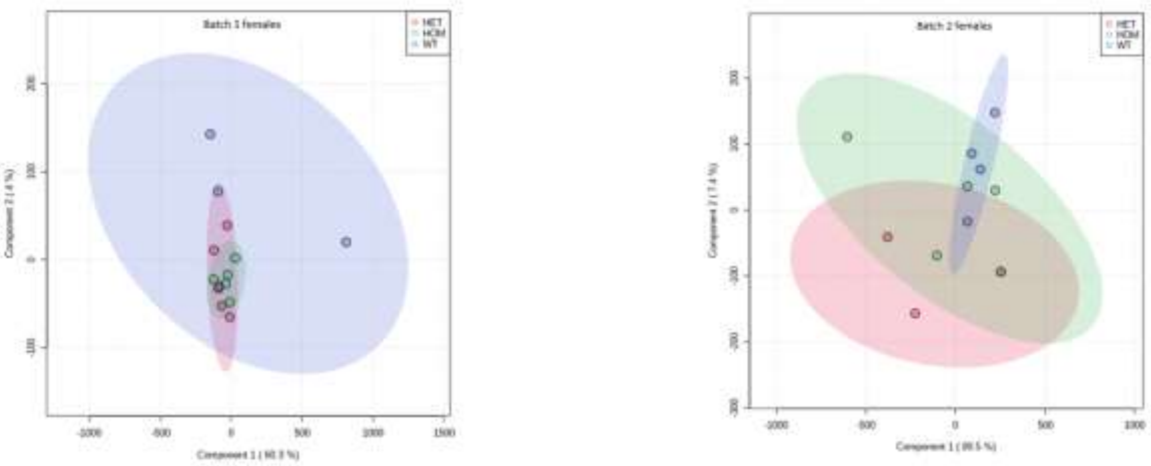

Figure 4. PCA analysis for females Genotype in batch 1 and batch 2

However, batch 1 females indicated an overlap between genotypes despite of close clustering of samples while batch 2 females indicated somewhat separation between genotypes despite of samples been spread out in the group (Figure 4). Hence, it is hard to speculate whether it was the close clustering that indicated variability in males in batch 1. It could be the difference between females in batch 2 indicating variation strong enough that was detected by PCA despite the samples within batch 2 been spread out. However, genotypes within each gender within batch 1 and batch 2 were not significant. Again, due to small sample size which results in a low statistical power the significance indicated might not provide important information.